Anthony Katona

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Summary Statement

Chemistry PhD specializing in polymer and ceramic coating processes, photothermal materials processing, and materials characterization. Experienced in lab-scale process development, class IV laser systems, safety oversight, and custom software and electronics development for productivity advancement. Skilled at communication to a wide range of knowledge levels and building processes that improve transparency and maintainability.

Experience & Education

The Pennsylvania State University *2023–current*
Chemistry (State College, PA) Postdoctoral Researcher

- led laser-processing development for UHTC preceramic polymer coating precursors within an 8-group MURI collaboration, establishing experimental procedures, air-free workflows, and process-parameter screening methods
- built remote monitoring instrumentation and a transparent flow/purge chamber for air-sensitive sample processing, improving real-time observation and control during laser treatment
- designed nanosecond pyrometer to inform collaborating theory groups of accessible peak temperatures during laser pulses
- integrated containerization into multiple existing project software pipelines with complex dependency stacks to facilitate access to tools between collaborators

The Pennsylvania State University *2023*
Chemistry (State College, PA) PhD

- developed photo-thermal polymer techniques for coatings and 3D-printing applications
- modeled and printed custom 3DP components for managing optics and improving in-house HFR videography and contact angle goniometry
- synthesized ultra-fast/nano-scale thermal transport materials
- developed Python analysis pipelines for the conversion of profilometry, mechanical testing, DMA, and thermal instrument output to reproducible metrics and plots
- oversaw lab safety as both General and Laser Safety Officer, including management of chemical inventory and coordinating laboratory audits
- managed supply and maintenance of pumps, gas cylinders, and gloveboxes
- taught as TA for all 5 available first-year chemistry courses for a total of 8 semesters
- co-developed and implemented novel thermal conductivity undergraduate lab experiments for *Dr. Bratoljub Milosavljevic*'s physical chemistry course
- served as graduate student TA mentor for 1/3 of class and research mentor for 5 undergraduate researchers, teaching experimental design, instrument use, scripting for data analysis, and scientific writing

Susquehanna University *2015*
Biochemistry; Physics minor (Selinsgrove, PA) BS

- synthesized transition metal Janus complexes with *Dr. William Doughtery* as a thesis project
- assisted in galaxy classification model training with *Dr. Violet Mager* during summer research

Technical Skills

spectroscopy/microscopy: NMR, IR, UV-Vis, AFM, TEM (STEM, EDS)

materials: mechanical (UTM, DMA) and thermal (DSC, TGA, TPS) tests; lathes, mills, 3DP, engravers

laser: class IV CW/pulsed safety, handling, and optics

electronics: circuit design, signal analysis, microcontrollers, LabVIEW

data/automation: Python (Pandas/NumPy/SciPy), C#, C++, LabVIEW, HTML, CSS, JS, SQL, ImageJ, WEKA segmentation, Docker

technical communication: LaTeX, markdown, Adobe CC (Ae, Il, Ps, Pt, Pr), Veusz, FL Studio, Blender 3D, Unreal Engine

Publications & Patents

publications:

- S Phillips *et al.*, 2026 review. Indust. Chem. & Mat., **"Nanometer-Scale Surface Roughness..."**
- A Katona & B Lear, 2025. J. Phys. Chem. C, **"Photothermal Curing of Polydimethylsiloxane: Carbon Black Composites Results in Changes to Polymer Topography, Cross-Link Density, and Mass Density"**
- A Blasone *et al.*, 2025. **"Rapid Photothermal Curing of PDMS on Paper"**
- A Katona & B Lear, 2024. Macromol., **"Effective photothermal curing of PDMS using ultra-low loadings of carbon black"**
- V Mager *et al.*, 2018. ApJ 864, **"Galaxy Structure in the Ultraviolet..."**

patent filings:

- co-inventor, "Photothermal Generation of Ceramics," U.S. Provisional Patent Application No. 63/910,435, filed Nov. 3, 2025
- co-inventor, "Photothermal Curing of Polymers on Thermally Sensitive Substrates," U.S. patent application pending; priority applications filed July, November, and December 2024

Outreach & Volunteer Work

Exploration-U *2019*

- co-presented polymer science demonstration at Bellefonte High School

Graduate Women in Science *2017–2018*

- led demonstration on quantum locking in superconductors for high school students
- designed and led a demonstration on the physics of sound for Girl Scouts
- designed and led a demonstration on the psychology of bias for Girl Scouts

Pennsylvania Junior Academy of Science *2017*

- served as a judge for student science presentations at the Penn State regional competition.